Flight Price Prediction Using Linear Regression

R-based Machine Learning Project

Submitted by: Hiba Muhammed | Roll No: 25 | S6IE
Department of Information Engineering | 2025-2026

Abstract

This project predicts airline ticket prices using Linear Regression in R. The dataset has 300,153 records with 11 features. After IQR outlier removal, two models were built: M1 (all features, R2=0.9129, RMSE=INR 6,664) and M2 (top 5 features, R2=0.9021). Seat class, stops, and airline are the strongest predictors.

1. Introduction

Airline ticket prices are dynamic and influenced by airline, route, stops, booking time, seat class, and departure schedule. This project applies Linear Regression to model these patterns and predict ticket prices accurately using R.

2. Dataset Description

The dataset contains 300,153 rows and 11 columns. Target variable is price (INR):

Feature	Type	Description
airline	Categorical	6 airlines (SpiceJet, Vistara, AirAsia, Indigo, GO_FIRST, Air_India)
source_city	Categorical	Departure city (6 cities)
departure_time	Categorical	Time bin: Early Morning, Morning, Afternoon, Evening, Night, Late Night
stops	Categorical	zero, one, two_or_more
arrival_time	Categorical	Arrival time bin (6 categories)
destination_city	Categorical	Destination city (6 cities)
class	Categorical	Economy or Business
duration	Numeric	Flight duration in hours
days_left	Numeric	Days between booking and departure
price	Target	Ticket price in INR

3. Exploratory Data Analysis

3.1 Price by Airline (Delhi to Mumbai)

Vistara and Air India charge premium fares; budget carriers like SpiceJet and AirAsia offer lower prices on the same route.

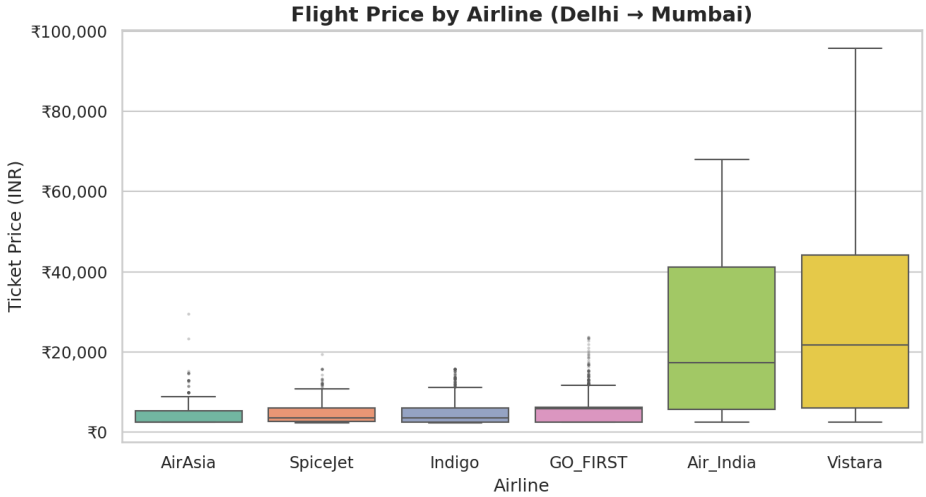

Figure 1: Ticket price by airline for Delhi to Mumbai route.

3.2 Price vs. Booking Time

Last-minute tickets (1-2 days before departure) cost significantly more. Booking 30+ days early yields the lowest prices.

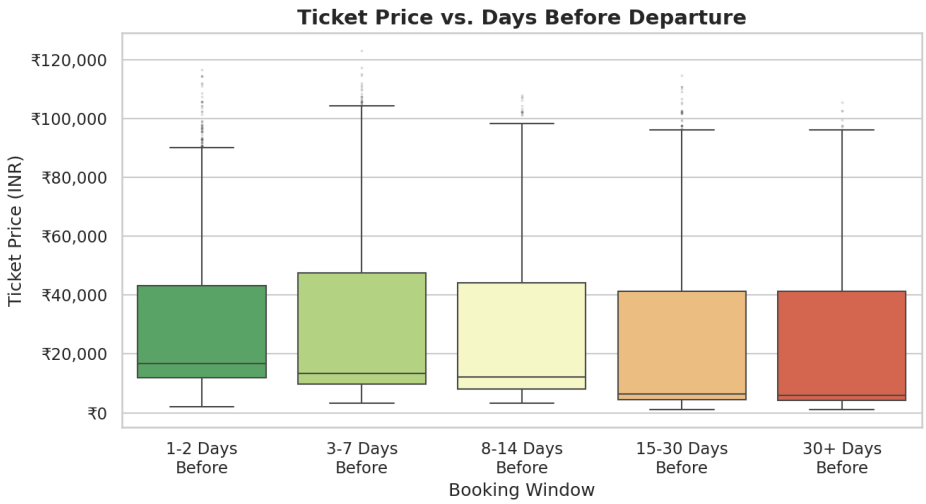

Figure 2: Ticket prices across booking windows.

3.3 Price by Departure and Arrival Time

Late-night and early-morning flights are cheaper. Daytime flights, especially Morning to Afternoon, command higher prices.

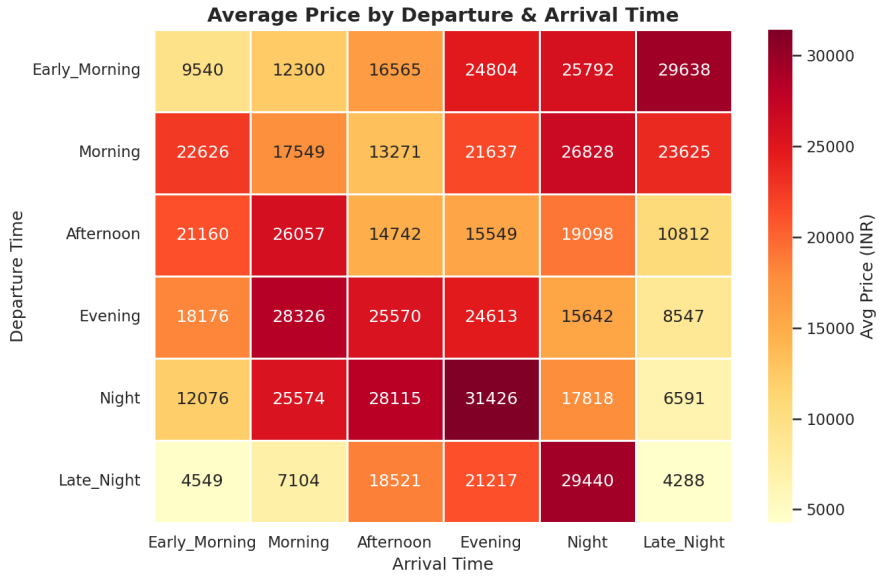

Figure 3: Heatmap of average price by departure and arrival time.

3.4 Price Distribution and Outlier Removal

IQR-based outlier removal eliminated 123 records, leaving 300,030 clean records.

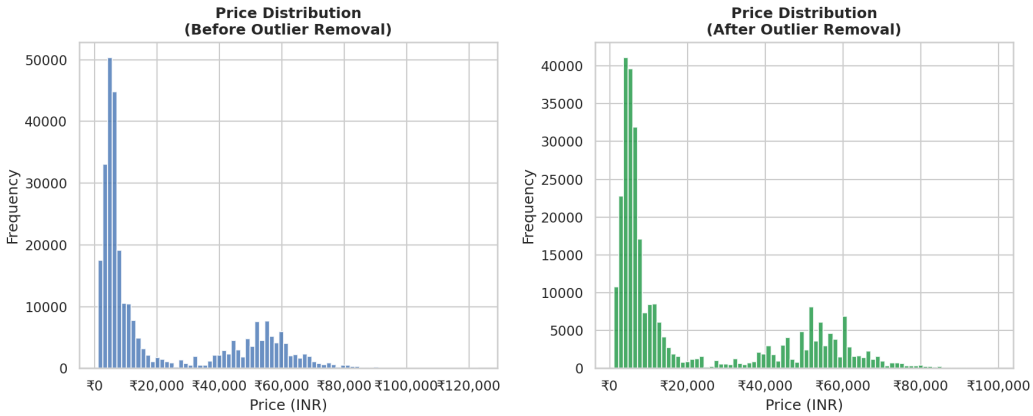

Figure 4: Price distribution before (left) and after (right) outlier removal.

4. Methodology

Preprocessing: Dropped the high-cardinality flight column. One-hot encoded all categorical features (drop-first).

Train/Test Split: 80% training, 20% testing (seed=42).

M1: Linear Regression trained on all encoded features.

Feature Selection: Top 5 features by absolute coefficient magnitude from M1.

M2: Linear Regression retrained on top 5 features only.

Evaluation: R2 and RMSE on the held-out test set.

5. Results

5.1 Model Comparison: M1 vs M2

Metric	M1 – All Features	M2 – Top 5 Features
R2	0.9129	0.9021
RMSE	INR 6,664.38	INR 7,066.93
Features	All encoded features	5 features

M1 explains ~91% of price variance. M2 with only 5 features achieves R2=0.9021, showing a small feature set captures most predictive power.

5.2 Feature Importance

Top 5 features: class_Economy, stops_zero, airline_Vistara, airline_SpiceJet, airline_Indigo. Seat class and stops dominate pricing.

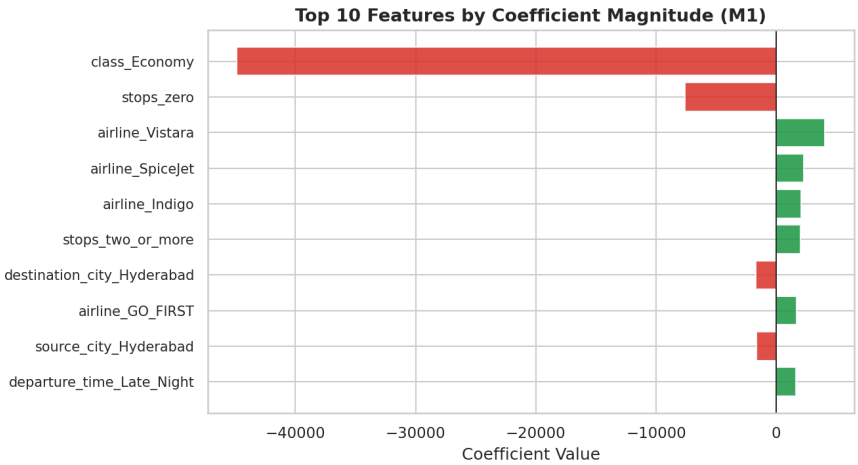

Figure 5: Top 10 features by regression coefficient magnitude.

5.3 Actual vs. Predicted Prices

Points cluster tightly around the diagonal (perfect fit line), confirming strong model accuracy.

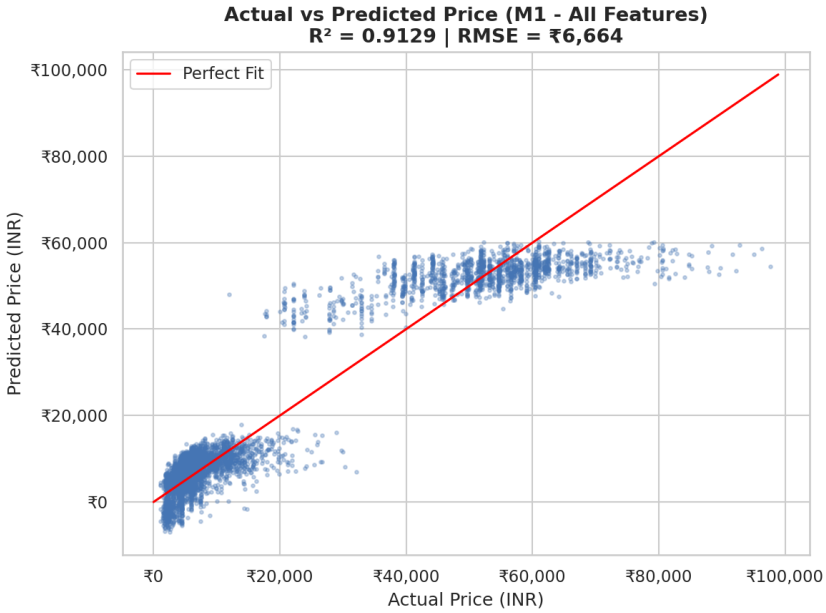

Figure 6: Actual vs. Predicted prices for Model M1.

6. Conclusion

Linear Regression achieved R2=0.9129 and RMSE=INR 6,664. Seat class, stops, and airline are the strongest price predictors. Last-minute bookings cost more. M2 with 5 features retains 90%+ accuracy, confirming a parsimonious model is viable.

Future work: ensemble methods (Random Forest, Gradient Boosting), real-time data, and a web application for passenger price estimation.

7. References

1. R Core Team (2024). R: A Language and Environment for Statistical Computing.
2. Wickham H. et al. (2019). Welcome to the tidyverse. JOSS.
3. Kuhn M. (2008). Building predictive models in R using the caret package. JSS.
4. Kaggle Flight Price Prediction Dataset.